Smart Urban Mobility and Traffic Intelligence Dashboard

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Project Title	Smart Urban Mobility and Traffic Intelligence Dashboard
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1. Introduction

The project develops a Power BI solution for analysing urban mobility, traffic conditions and ride demand. The work is organized into three milestones. Milestone 1 establishes the ride dataset and analytical data model. Milestone 2 focuses on traffic volume, speed, congestion, accidents, weather filtering and geographic traffic conditions. Milestone 3 focuses on ride demand, time-of-day behaviour, surge pricing, weather impact and clearly labelled scenario analysis for alternative mobility.

2. Tools Used

- Power BI – dashboard development and interactive visualization
- Power Query – data preparation and transformation
- DAX – KPI measures and analytical calculations
- Microsoft Excel – supporting data inspection and preparation
- Data Modelling – star-schema design and relationships
- Data Visualization – presenting mobility and traffic patterns

3. Project Flow

The project follows a progressive analytical flow: Ride Dataset Preparation and Modelling → Geospatial and Traffic Analytics → Demand Intelligence and Mobility Analytics. The same structured data model supports the later dashboards.

4. Milestone 1 – Ride Dataset Preparation and Data Modelling

4.1 Ride Dataset Preparation

The main source used for the project is the ride_safety_dataset. It contains ride, traffic, weather, location and safety-related fields used across the project. Power Query was used to inspect the available fields, review data types and structure, and organize the source data for Power BI analysis. This preparation provides the base for the measures and visual analysis developed in the later milestones.

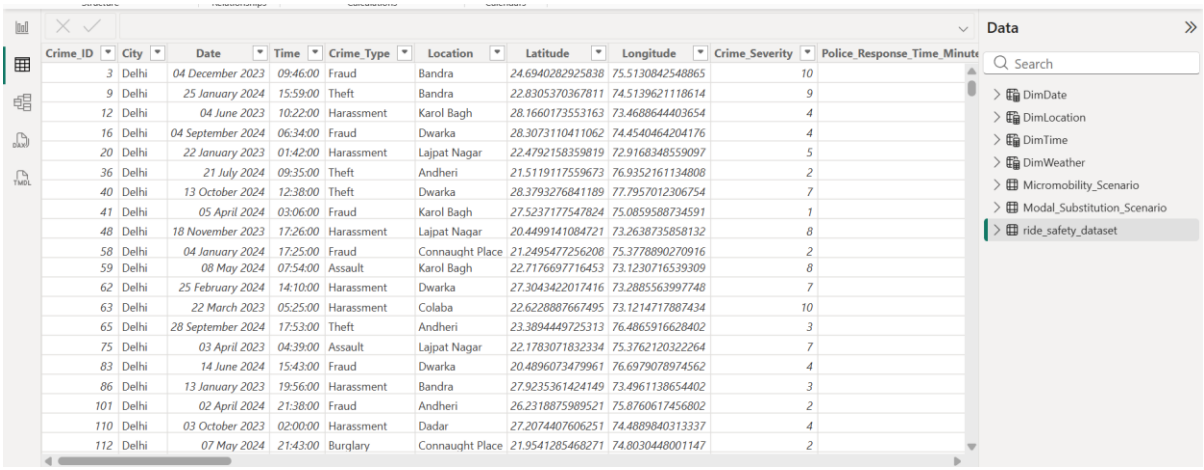
4.2 Star-Schema Model

The ride_safety_dataset is used as the central fact table in a star-schema model. Four supporting dimension tables were created: DimDate, DimLocation, DimWeather and DimTime. Each dimension is related to the central fact table and provides descriptive fields for filtering and grouping ride and mobility measures. The model provides a consistent analytical foundation for the Milestone 2 and Milestone 3 dashboards.

4.3 Milestone 1 Data Foundation Explanation

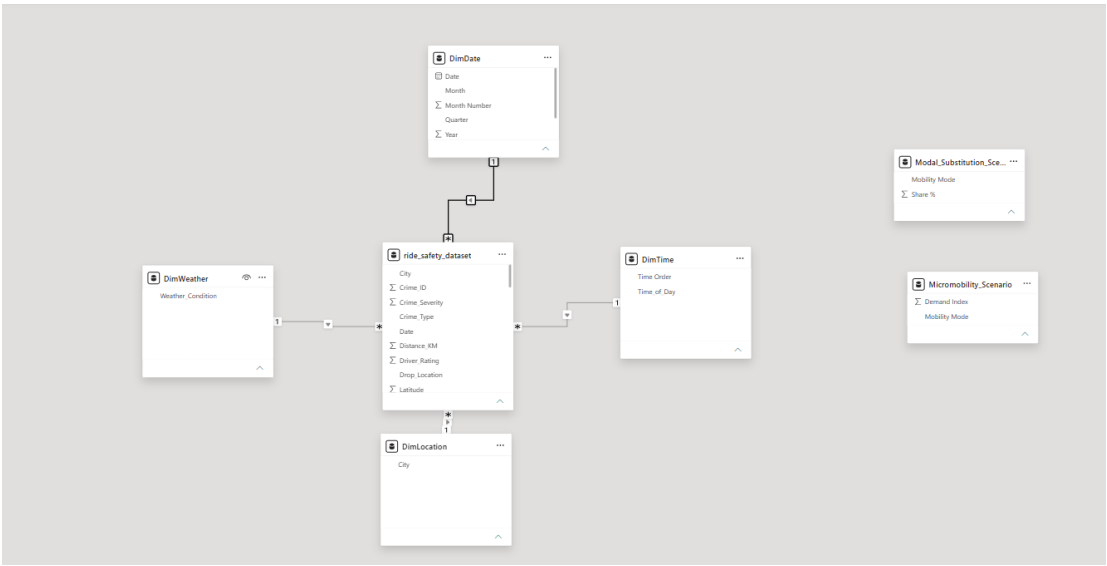
Milestone 1 establishes the technical foundation for the complete project. The dataset view shows the detailed source records available for analysis, while the model view shows how the central ride data is organized and connected to the Date, Location, Weather and Time dimensions. This structure supports consistent filtering, aggregation and DAX calculations in the later dashboards.

Ride Dataset – ride_safety_dataset



Crime_ID	City	Date	Time	Crime_Type	Location	Latitude	Longitude	Crime_Severity	Police_Response_Time_Minute
3	Delhi	04 December 2023	09:46:00	Fraud	Bandra	24.6940282925838	75.5130842548865	10	
9	Delhi	25 January 2024	15:59:00	Theft	Bandra	22.8305370367811	74.5139621118614	9	
12	Delhi	04 June 2023	10:22:00	Harassment	Karol Bagh	28.1660173553163	73.4688644403654	4	
16	Delhi	04 September 2024	06:34:00	Fraud	Dwarka	28.3073110411062	74.4540464204176	4	
20	Delhi	22 January 2023	01:42:00	Harassment	Lajpat Nagar	22.47921583559819	72.9168348559097	5	
36	Delhi	21 July 2024	09:35:00	Theft	Andheri	21.5119117559673	76.9352161134808	2	
40	Delhi	13 October 2024	12:38:00	Theft	Dwarka	28.3793276841189	77.7957012306754	7	
41	Delhi	05 April 2024	03:06:00	Fraud	Karol Bagh	27.5237177547824	75.0859588734591	1	
48	Delhi	18 November 2023	17:26:00	Harassment	Lajpat Nagar	20.4499141084721	73.2638735858132	8	
58	Delhi	04 January 2024	17:25:00	Fraud	Connaught Place	21.2495477256208	75.3778890270916	2	
59	Delhi	08 May 2024	07:54:00	Assault	Karol Bagh	22.7176697716453	73.1230716539309	8	
62	Delhi	25 February 2024	14:10:00	Harassment	Dwarka	27.3043422017416	73.2885563997748	7	
63	Delhi	22 March 2023	05:25:00	Harassment	Colaba	22.6228887667495	73.1214717887434	10	
65	Delhi	28 September 2024	17:53:00	Theft	Andheri	23.3894449725313	76.4865916628402	3	
75	Delhi	03 April 2023	04:39:00	Assault	Lajpat Nagar	22.1783071832334	75.3762120322264	7	
83	Delhi	14 June 2024	15:43:00	Fraud	Dwarka	20.4896073479961	76.6979078974562	4	
86	Delhi	13 January 2023	19:56:00	Harassment	Bandra	27.9235361424149	73.4961138654402	3	
101	Delhi	02 April 2024	21:38:00	Fraud	Andheri	26.2318875989521	75.8760617456802	2	
110	Delhi	03 October 2023	02:00:00	Harassment	Dadar	27.2074407606251	74.4889840313337	4	
112	Delhi	07 May 2024	21:43:00	Burglary	Connaught Place	21.9541285468271	74.8030448001147	2	

Power BI Model View – Star Schema



5. Milestone 2 – Geospatial and Traffic Analytics

5.1 Traffic Volume and Speed Analysis

The Traffic Volume Trend visual shows how recorded traffic activity changes over time. Average Speed is presented as a KPI and summarizes the average recorded traffic speed. Together, these visuals provide a time-based view of traffic activity and movement conditions.

5.2 Congestion and Traffic Condition Analysis

Average Congestion provides an overall congestion indicator. Traffic Condition Distribution groups the recorded congestion levels into Low, Medium, High and Very High categories. The Traffic Density & Speed visual then compares these categories with average speed to examine how speed varies across traffic conditions.

5.3 Geographic Traffic Analysis

The Geographic Traffic Overview uses latitude and longitude to plot the recorded observations on a map. This provides a spatial view of traffic and mobility conditions and allows the geographic distribution of observations to be examined together with city and traffic-related information.

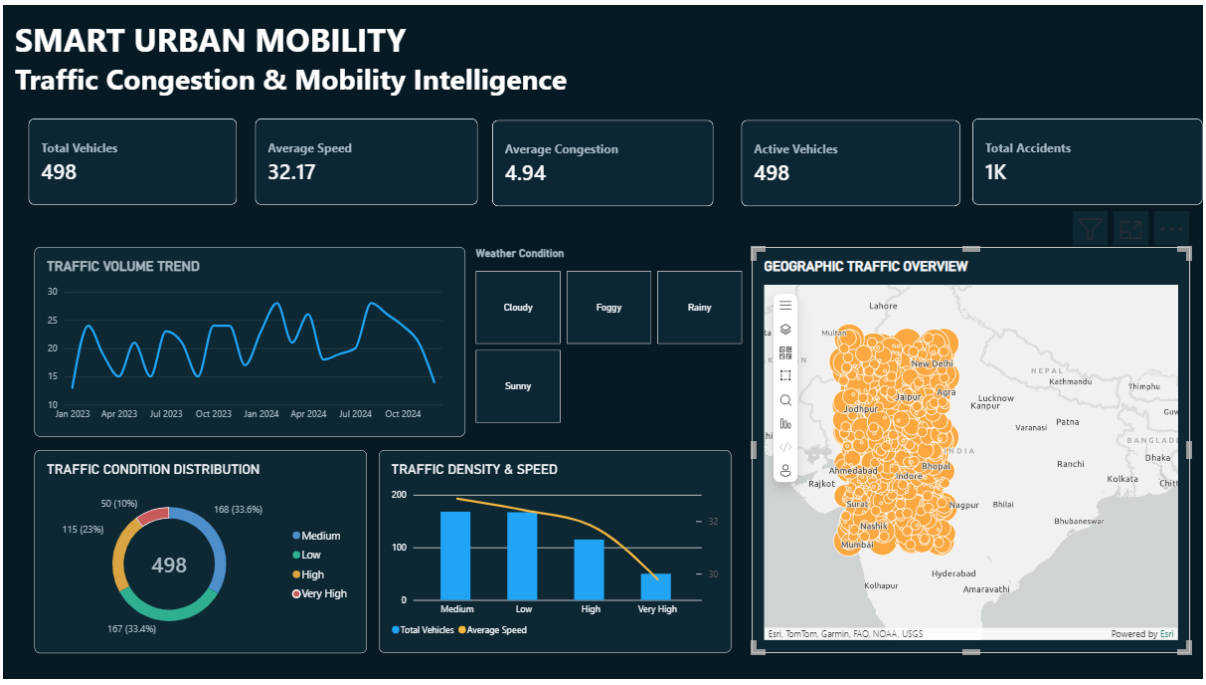
5.4 Weather and Accident Indicators

Weather Condition is available as an interactive filter so that the traffic dashboard can be viewed for selected weather categories. Total Accidents summarizes the reported accident values represented in the dataset. The Active Vehicles KPI uses distinct Ride_ID records as a proxy for active mobility records.

5.5 Milestone 2 Dashboard Explanation

Milestone 2 converts the prepared data model into a traffic intelligence dashboard. The dashboard combines five summary indicators—Total Vehicles, Average Speed, Average Congestion, Active Vehicles and Total Accidents—with a traffic trend, weather filter, geographic traffic map, traffic-condition distribution and traffic-density/speed comparison. This milestone provides both temporal and geographic views of the traffic conditions represented in the dataset.

Milestone 2 Dashboard – Traffic Congestion & Mobility Intelligence



6. Milestone 3 – Demand Intelligence and Mobility Analytics

Milestone 3 Dashboard – Demand Intelligence & Mobility Analytics



6.1 Demand Overview and KPI Analysis

The dashboard begins with five ride-demand and service indicators: Total Rides (498), Average Fare (834.18), Average Ride Duration (61.62 minutes), Driver Rating (2.91) and Ride Demand (500). Total Rides represents distinct Ride_ID records, while Ride Demand represents the total ride records counted in the dataset. These KPIs provide the overall context for the detailed demand analysis.

6.2 Ride Demand Trend

The Ride Demand Trend visual shows how recorded ride demand changes across the available months. It provides a time-based view of fluctuations in ride activity and helps identify periods with relatively higher or lower demand.

6.3 Demand by Time of Day

The Ride Demand Distribution across Time Periods compares demand during Morning, Afternoon, Night and Evening. The current dashboard records 118, 136, 132 and 114 respectively, allowing ride activity to be compared across different periods of the day.

6.4 Surge Pricing Impact

The Surge Pricing Impact visual combines ride-record volume and average ride fare across different surge-pricing factors. The columns represent the number of ride records and the line represents average fare, allowing the relationship between surge level, ride activity and fare to be examined within the available data.

6.5 Weather Impact on Ride Demand

The Weather Impact on Ride Demand donut compares recorded ride demand across Cloudy, Sunny, Foggy and Rainy conditions. The current dashboard shows 117 Cloudy records, 131 Sunny records, 118 Foggy records and 134 Rainy records. The visual is used to examine how recorded ride activity varies across weather conditions.

6.6 Micromobility Demand Index – Scenario

The Micromobility Demand Index compares Bicycle, E-Scooter, E-Bike and Walking using scenario index values of 18, 14, 11 and 7. The source ride dataset does not contain an observed micromobility-demand field, so these values are presented explicitly as a scenario rather than as measured demand.

6.7 Modal Substitution – Scenario

The Modal Substitution visual compares Ride-Hailing, Public Transit, Cycling and Walking using scenario shares of 58%, 24%, 12% and 6%. These values demonstrate how alternative modes can be represented for mobility planning; they are not observed modal-substitution measurements from the source ride dataset.

6.8 Milestone 3 Outcome

Milestone 3 extends the traffic analysis from Milestone 2 into ride-demand intelligence. It combines overall ride KPIs, demand trend, time-of-day demand, surge pricing and weather impact, and then adds clearly labelled scenario analysis for micromobility and modal substitution. This provides a broader view of ride activity and demonstrates how alternative mobility patterns can be represented without treating scenario inputs as observed data.

7. Dashboard Final Output

Complete Final Dashboard Output



The final dashboard presents the completed Milestone 3 demand-intelligence view as the consolidated project output. It brings together the ride KPIs, demand trend, time-of-day analysis, surge pricing, weather impact, micromobility scenario and modal-substitution scenario in one interactive Power BI page.

8. Conclusion

The Smart Urban Mobility and Traffic Intelligence Dashboard demonstrates a structured Power BI workflow from ride dataset preparation and star-schema modelling to traffic and demand analysis. Milestone 1 establishes the data and modelling foundation, Milestone 2 analyses traffic and geographic conditions, and Milestone 3 analyses ride demand and mobility behaviour. Together, the dashboards provide a connected view of urban mobility while clearly distinguishing measured dataset analysis from scenario-based mobility analysis.

9. References

- Microsoft Power BI Documentation
- Microsoft Excel Documentation
- Power BI Learning Resources
- Kaggle / source dataset: ride_safety_dataset.csv